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Investigating Balanced Warm-Up and Gradual Unfreezing for Deep Learning-Based Mammography Classification



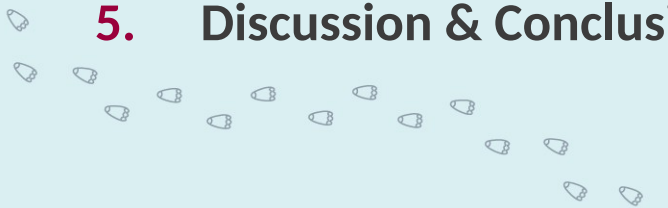
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Graduate Defense · Supervisor: Prof. Tad Gonsalves

An exploratory comparison of fine-tuning strategies, not a clinically deployable model.

Presentation roadmap

1. Background & Prior Work
2. Strategy Design
3. Research Setting
4. Results
5. Discussion & Conclusion



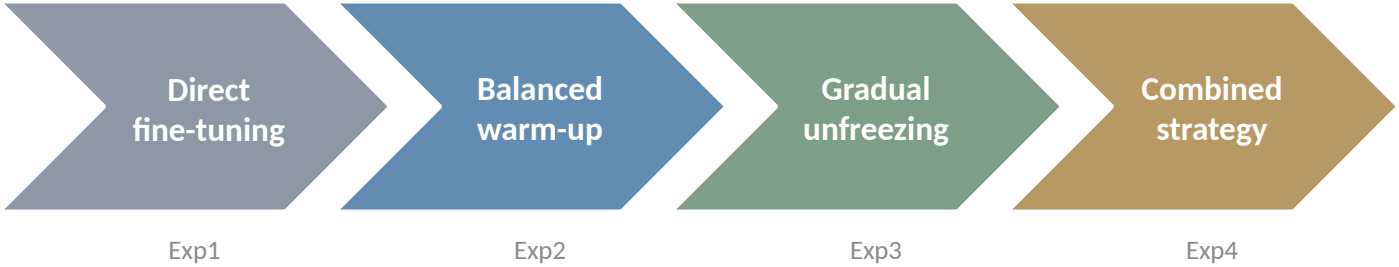
1. **Background & Prior Work**
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Four fine-tuning strategies in one controlled comparison

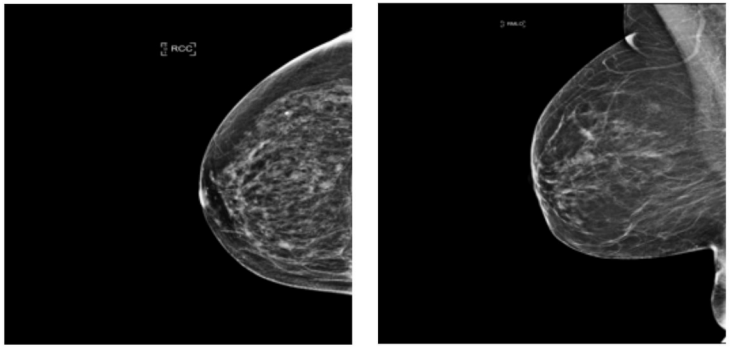
OVERVIEW · STUDY MAP

How should an ImageNet-pretrained CNN be fine-tuned for highly imbalanced mammography classification?



Mammography screening creates a rare-positive learning problem

BACKGROUND · TASK CONTEXT



Example mammographic view

- Breast cancer is the most common cancer among women worldwide and caused about 670,000 deaths in 2022. Early detection is strongly associated with better prognosis, regular imaging-based screening is essential.
- Mammography is the most widely used tool for breast cancer screening.
- Deep learning has shown strong potential across medical imaging tasks.
- But real screening data are hard to train on: rare positives, subtle lesions.

- **The difficulty is to make deep learning model learn useful signals from rare positive cases.**

Two training challenges define the study

MOTIVATION · TRAINING CHALLENGES

CHALLENGE 1

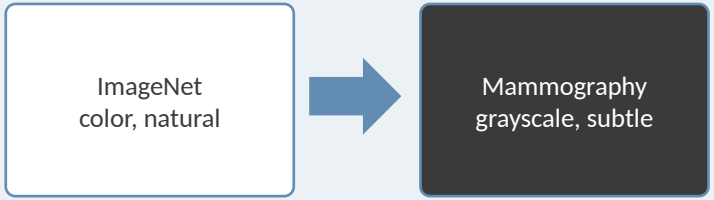
Severe class imbalance



~2% of samples are positive, so early training is dominated by negatives.

CHALLENGE 2

Transfer-learning domain gap



Low-contrast, grayscale images with small, subtle lesions.

- **The real question is, under severe imbalance and domain gap, how to adapt a pretrained model to the specific task.**

Mammography DL has advanced through performance-led model design

RELATED WORK · MAINSTREAM FOCUS

Shen et al. 2019	End-to-end mammography model; after lesion-patch pretraining, the whole-image classifier can be trained with image-level labels. Reported DDSM per-image AUC 0.88–0.91. (Sci. Rep.)
McKinney et al. 2020	Evaluated an AI screening system on UK and US screening cohorts; reduced both false positives and false negatives and provided evidence of cross-population generalization. (Nature)
Wu et al. 2020	A deep CNN trained on over 200,000 exams and one million images reached breast-level AUC 0.895; a reader study showed the model and radiologists are complementary. (IEEE TMI)

Mammography DL has shown strong potential for breast cancer screening, classification, and detection;
Mainly aims to improve predictive performance and clinical utility

Performance gains come from mammography-specific modeling

RELATED WORK · MAINSTREAM METHODS

Shen et al. 2019	Annotation-efficient end-to-end pipeline: lesion-patch pretraining → whole-image classification with image-level labels. (Sci. Rep.)
Lotter et al. 2021	Annotation-efficient robust detection system; extended from mammography to DBT and tested for generalization. (Nat. Med.)
Wu et al. 2020	Large-scale exam-level model built on two-stage patch-level and breast-level training. (IEEE TMI)
Shen et al. 2021	Global-local architecture with weakly supervised localization from image-level labels. (MedIA)

Mainstream mammography DL advances are mainly driven by task-specific model, pipeline, input, localization, and scale design. Fine-tuning may be used, but it is usually not the core research target.

Fine-tuning strategy changes medical imaging outcomes

RELATED WORK · EVIDENCE IT MATTERS

Davila et al. 2024

Compared eight fine-tuning strategies across three CNN architectures (ResNet-50, DenseNet-121, VGG-19) and five medical imaging modalities: X-ray, MRI, histology, dermoscopy, and endoscopy. Linear probing followed by full fine-tuning improved results in over half of the cases, and an adaptive learning-rate schedule (Auto-RGN) gave up to 11%. (IVC 2024)

Broader medical imaging studies indicate that fine-tuning strategy can affect model performance and should be studied as an experimental variable.

Research focus: fine-tuning strategy under mammography imbalance

RELATED WORK · RESEARCH GAP

- Mammography DL has largely advanced through performance-driven model and system development.
- Fine-tuning strategy remains less explicitly examined as a central research object in this task.
- Broader medical imaging studies suggest that fine-tuning choices can affect outcomes.

- **Therefore, this study makes fine-tuning strategy the central object.**

Controlled question

Under a fixed ResNet101, patient ID + laterality groups, and unified metrics, how do initialization-oriented warm-up and adaptation-oriented gradual unfreezing affect learning behavior under severe class imbalance?

1. Background & Prior Work

2. **Strategy Design**

3. Research Setting

4. Results

5. Discussion & Conclusion

Training challenges translate into strategy roles

STRATEGY DESIGN · OVERVIEW

Challenge	Training issue	Strategy
Class imbalance	Early training dominated by negatives	Balanced warm-up (initialization)
Domain gap	Updating all layers at once may be unstable	Gradual unfreezing (adaptation)
Both	Initialization and adaptation both matter	Combined

Warm-up = *initialization* **Gradual unfreezing = *adaptation***

Balanced warm-up tests initialization under imbalance

STRATEGY DESIGN · INITIALIZATION

This is NOT learning-rate warm-up.



PRINCIPLE

Inspired by resampling-based imbalance handling.

HYPOTHESIS

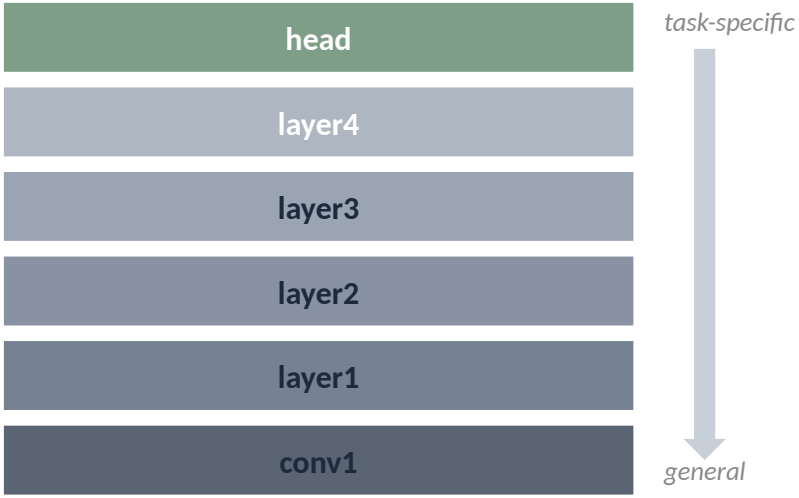
Better early initialization may help full-data fine-tuning.

RISK

The subset benefit may not transfer stably.

Gradual unfreezing tests controlled adaptation

STRATEGY DESIGN · ADAPTATION



- **Train the head first, then unfreeze deeper layers step by step: controlled adaptation.**

FEATURE HIERARCHY

Lower layers learn general features; higher layers are more task-specific.

PROCEDURE

Unfreeze in order: head → layer4 → layer3 → layer2 → layer1 → conv1.

HYPOTHESIS

Gradual unfreezing gives controlled adaptation.

1. Background & Prior Work
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Three strategy questions structure the experiments

RESEARCH QUESTIONS · EXPERIMENT MAP

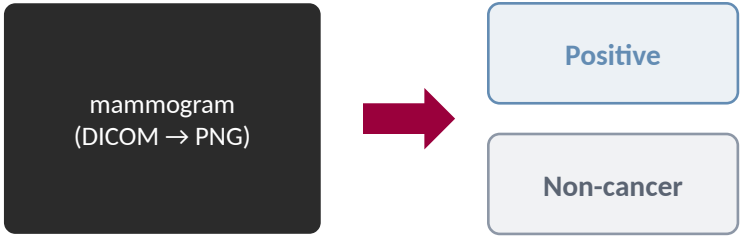
RQ1 · Warm-up	Can balanced warm-up improve early learning and carry over to full imbalanced training?
RQ2 · Unfreezing	Can gradual unfreezing improve performance and stability?
RQ3 · Combined	Can combining the two give a better performance-stability trade-off?

- **Each strategy question maps to one main experiment, with Exp1 as the baseline.**

Binary Classification under Severe Imbalance

RESEARCH SETTING · DATASET

Image → Positive / Non-cancer



RSNA Screening Mammography Breast Cancer Detection dataset

Split	Positive	Negative	Positive ratio
Full training	936	42,792	2.14%
Validation	222	10,756	2.02%
Subset A (1:3)	936	2,767	25.28%

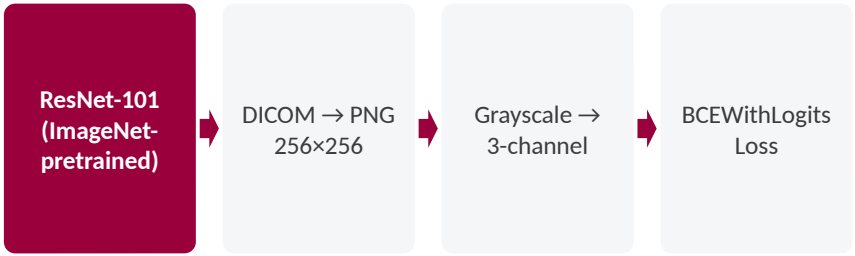
Only ~2% of the full dataset is positive: the number to remember for the rest of the talk.

Split integrity: subsets are drawn from training only; validation is untouched; no breast-group overlap.

- **~2% positive ratio: this defines the central training challenge.**

Fixed model and metrics isolate training-strategy effects

RESEARCH SETTING · MODEL & EVALUATION



Everything above is held constant across all four experiments.

ROC-AUC
overall ranking quality

PR-AUC
positive-class ranking

Validation loss
training stability

- **Accuracy is misleading at 2% positives: a model can score 98% by predicting “negative” every time.**

Four experiments isolate individual and combined effects

RESEARCH SETTING · EXPERIMENT MAP

Experiment	Design	Purpose
Exp1	Full-data fine-tuning	Baseline
Exp2	1:3 warm-up → full data	Warm-up alone
Exp3	Full-data gradual unfreezing	Unfreezing alone
Exp4	Warm-up → recalibration → gradual unfreezing	Combined

Full-data recalibration: readjusts the warm-up checkpoint to the original imbalanced distribution before gradual unfreezing.

- **One experiment per research question: this is the map for the results section.**

1. Background & Prior Work
2. Research Setting
3. Strategy Design
4. **Results**
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Learning behavior matters beyond peak ROC-AUC

↑ ROC-AUC = better · ↓ Val loss = better

RESULTS · READING GUIDE

1. Checkpoint result
2. Is the AUC gain sustained or transient?
3. Does val loss stay stable or rise?
4. Final interpretation

Experiment	ROC-AUC	Val loss	PR-AUC
Exp1	0.5780	0.4115	0.0242
Exp2b peak	0.6591	0.4341	0.0426
Exp3	0.6601	0.4062	0.0486
Exp4	0.6711	0.3980	0.0482

Not just "which number is biggest": is the improvement real, or a brief peak? how each strategy affects learning behavior

- **Exp2b and Exp3 reach almost the same peak AUC: their loss behavior is the real story.**

↑ ROC-AUC = better · ↓ Val loss = better

Direct fine-tuning establishes a weak baseline

RESULTS · EXP1

DIRECT FINE-TUNING

ROC-AUC

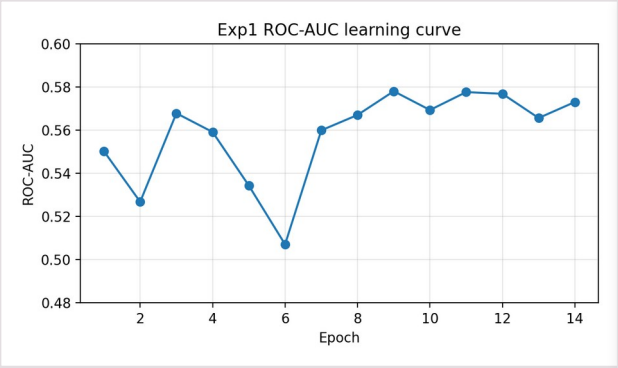
0.5780

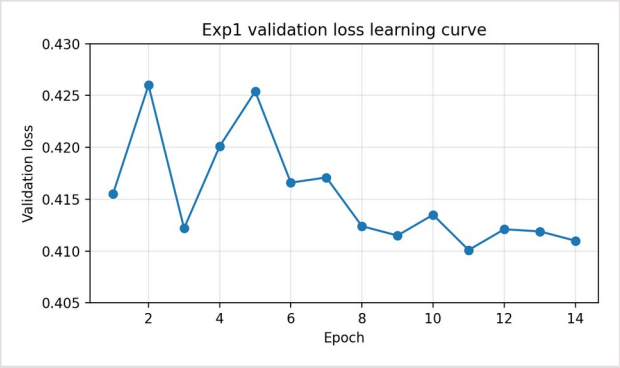
PR-AUC

0.0242

Val loss

0.4115





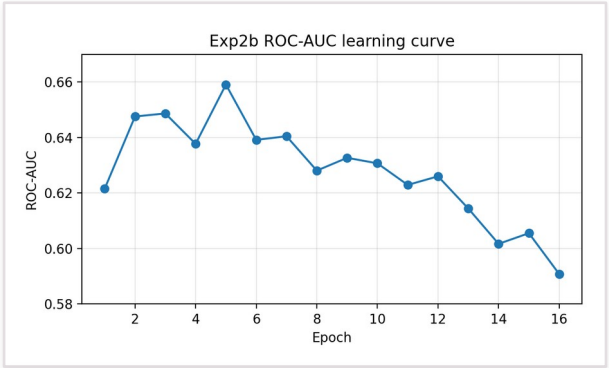
- **Direct fine-tuning shows limited positive-class learning.**

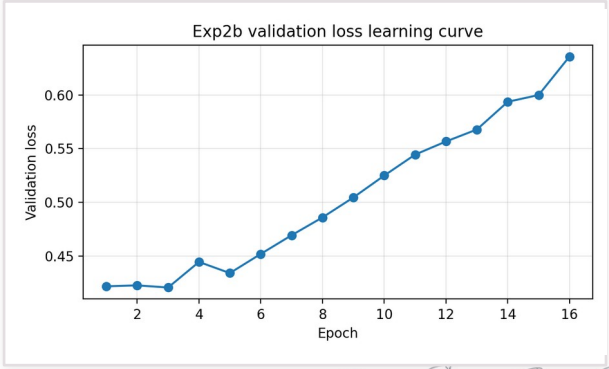
Warm-up improves early ranking but transfers unstably

RESULTS · EXP2

Result	ROC-AUC	PR-AUC	Val loss
Exp2a transition ckpt.	0.6436	0.0347	0.2644
Exp2b peak	**0.6591**	**0.0426**	**0.4341**
Exp2b loss-ctrl.	0.6487	0.0390	0.4207

1. *More balanced subset helps early ranking while keeping loss relatively low.*
2. *Part of the warm-up benefit carries over to full-data training.*
3. *The better-AUC checkpoint carries worse loss behavior.*

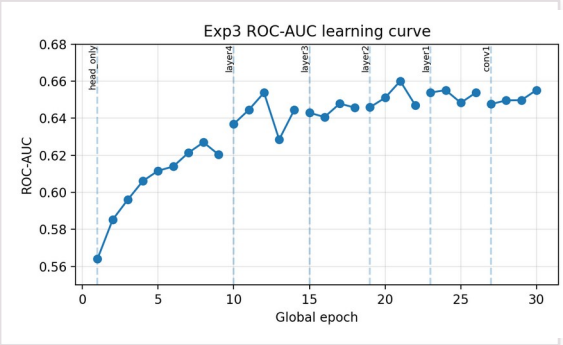


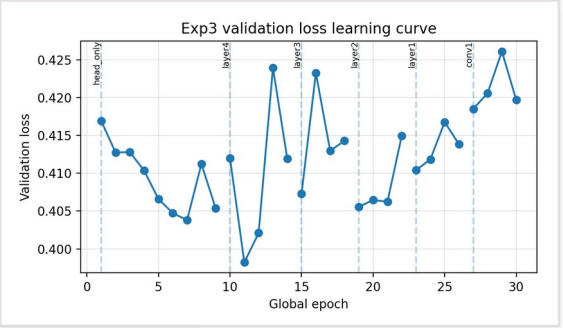


Gradual unfreezing improves stability at comparable AUC

RESULTS · EXP3

Method	ROC-AUC	Val loss	PR-AUC
Exp1 baseline	0.5780	0.4115	0.0242
Exp2b peak	0.6591	0.4341	0.0426
Exp3	**0.6601**	**0.4062**	**0.0486**





1. Gradual unfreezing clearly improves over direct fine-tuning.

2. AUC comparable to Exp2b, but lower val loss and higher PR-AUC.

- **Best result at layer2, not full unfreeze:**
  preserving lower-level representations may help retain transferable features

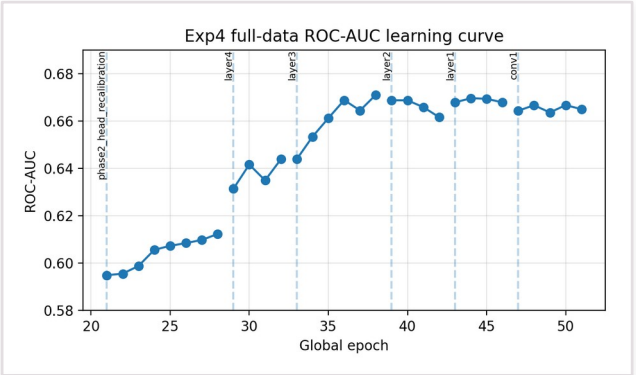
Combining strategies gives the best observed trade-off

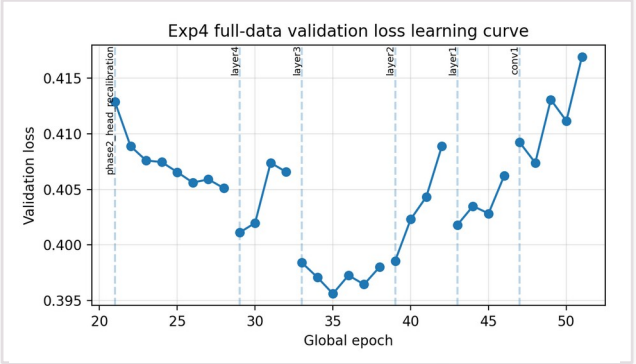
RESULTS · EXP4

Stage	ROC-AUC	PR-AUC	Val loss
Full-data recalib.	0.6123	0.0360	0.4051
Layer4	0.6439	0.0536	0.4066
Layer3 (best)	**0.6711**	**0.0482**	**0.3980**
Conv1	0.6667	0.0550	0.4111

1. Layer3 chosen as best AUC-loss trade-off.
PR-AUC is close across later stages (0.048–0.055).

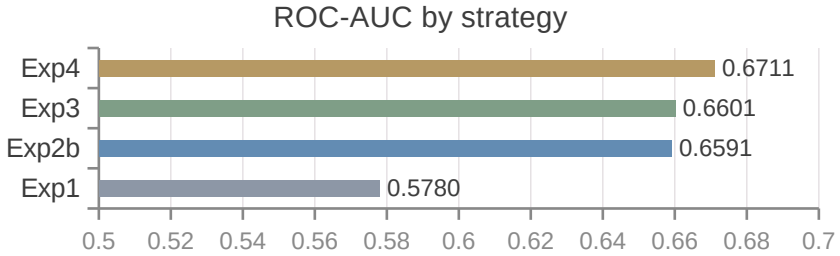
2. Balanced initialization and controlled adaptation may complement each other in this setting.

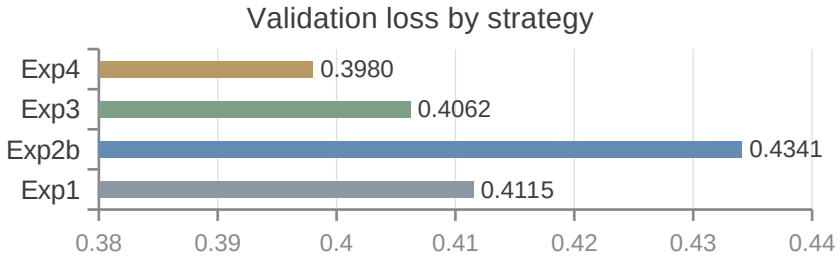




The cross-experiment pattern supports two strategy roles

RESULTS · SYNTHESIS





- Exp1: direct fine-tuning was weak
- Exp2: warm-up helped early learning, but transfer was unstable
- Exp3: gradual unfreezing gave more controlled adaptation
- Exp4: combined achieved the best AUC-loss trade-off

- **Both initialization and adaptation matter: combining them worked best in this setup.**
  Balanced warm-up helps the model start from a better point, but it is not stable enough by itself.
  Gradual unfreezing controls how the pretrained model adapts to mammography.

1. Background & Prior Work
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Research questions are answered through strategy behavior

DISCUSSION · RESEARCH QUESTIONS

RQ1

Warm-up

Improves early ranking;
transfer stability is limited.

RQ2

Unfreezing

Improves over baseline;
gives more controlled adaptation.

RQ3

Combined

Best observed AUC-loss trade-off;
interpret cautiously.

- **I read Exp4 as the best trade-off, not a proven, statistically significant win.**

Limitations define the next validation steps

DISCUSSION · LIMITATIONS & FUTURE WORK

LIMITATIONS

- Single-run experiments
- No statistical significance testing(bootstrap confidence intervals /Delong tests)
- Image-level only:

  No multi-view / ROI / ensemble

FUTURE WORK

- Multiple seeds
- Bootstrap CI / DeLong test
- Focal or class-balanced loss
- Higher-resolution input
- ROI-guided & multi-view models
- External validation

Controlled comparison reveals a consistent training pattern

CONCLUSION · KEY FINDINGS

- Direct fine-tuning is limited under severe imbalance.
- Warm-up improves early learning, but transfer alone is unstable.
- Gradual unfreezing gives more controlled adaptation.
- The combined strategy achieved the best observed AUC-loss trade-off.

Fine-tuning strategy affects learning behavior in this setting. Both initialization and adaptation matter.

This remains an exploratory comparison, not a clinically deployable model.

Training hyperparameters

APPENDIX

Item	Setting
Dataset	RSNA Screening Mammography
Split	Patient ID + laterality grouped split
Backbone	ImageNet-pretrained ResNet101
Input size	256 × 256
Preprocessing	DICOM → PNG, 3-channel, ImageNet norm.
Loss	BCEWithLogitsLoss
Optimizer	AdamW
Batch size	128 (Exp1); see configs for Exp2-4
Weight decay	5e-4 (Exp1); 1e-4 (Exp2-4)
Scheduler	ReduceLROnPlateau
Metrics	ROC-AUC, PR-AUC, validation loss

Experiment / phase	LR setting	Notes
Exp1 baseline	1e-3	Direct full-data fine-tuning
Exp2a warm-up	3e-5	1:3 subset warm-up
Exp2b transfer	1e-5	Full-data transfer after warm-up
Exp3 unfreezing	Layer-wise	Same policy as Exp4
Exp4 unfreezing	Layer-wise	Same policy as Exp3

Layer-wise LR policy *: larger LR for the classification head and higher layers; smaller LR for lower layers.*

- **Learning rates differed by training phase, but Exp3 and Exp4 used the same layer-wise LR policy during gradual unfreezing.**

Core Literature

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Thank you

I'm happy to take questions.

FOR OTHERS, WITH OTHERS

